

Week 1 Lab — Python Foundations

Course: Introduction to Big Data Analytics

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Part 0 · Warm-Up

Create a new notebook, rename it Week1_YourName



Part 1 · Variables & Data Types

Exercise 1.1

Create four variables about **yourself**: `my_name` (str), `my_age` (int), `my_gpa` (float — invent one), and `am_i_present` (bool). Print each variable together with its type.

Hint: `print(my_name, type(my_name))`

```
[10]: my_name = "Osama"
      my_age = 25
      my_gpa = 3.6
      am_i_present = True

      print(my_name, type(my_name))
      print(my_age, type(my_age))
      print(my_gpa, type(my_gpa))
      print(am_i_present, type(am_i_present))

Osama <class 'str'>
25 <class 'int'>
3.6 <class 'float'>
True <class 'bool'>
```

Exercise 1.2 — Type conversion

Real datasets often store numbers as text. Starting from the given line below, convert the value to an int, add 1 (a birthday!), and print the result.

`age_text = "21"` # given: a number trapped inside text

```
[11]: age_text = "21"
      age = int(age_text)

      age = age + 1

      print(age)

22
```

Question 1.3 — Think about it

What does "3" + "4" produce? Run it, then explain in one sentence why the answer is not 7.

```
[12]: print("3" + "4")

34
```

"3" + "4" produces "34" because both values are strings, so Python joins the text instead of adding the numbers.

Part 2 · Strings — Working with Text

Exercise 2.1 — f-strings

Using your variables from Exercise 1.1, print the following sentence with **one f-string**:

I am NAME, AGE years old, studying at AUCA.

Reminder: `f"Hello {my_name}"` — variables go inside `{ }`

```
[65]: my_name = "Osama"
      my_age = 25

      print (f"I am {my_name}, {my_age} years old , studying at AUCA.")

I am Osama, 25 years old , studying at AUCA.
```

Exercise 2.2 — String tools

Given `course = "Big Data Analytics"`, print: (a) the course name in UPPERCASE, (b) the number of characters it contains, (c) only the first 3 characters.

Hints: `.upper()` · `len(...)` · slicing `[0:3]`

```
[17]: course = "Big Data Analytics"

      print (course.upper())
      print (len(course))
      print (course[0:3])

BIG DATA ANALYTICS
18
Big
```

Part 3 · Lists — Your First Descriptive Statistics

Exercise 3.1

The quiz scores of your 5-person study group are given below. Print: (a) the first and the last score, (b) the number of scores, (c) the average, (d) the highest and the lowest score.

`scores = [72, 85, 91, 64, 78]`

Hints: `scores[0]` · `scores[-1]` · `len()` · `sum()/len()` · `max()` · `min()`

```
[66]: scores = [72, 85, 91, 64, 78]

print("First score:", scores[0])
print("Last score:", scores[-1])
print("Number of scores:", len(scores))
print("Average:", sum(scores) / len(scores))
print("Highest score:", max(scores))
print("Lowest score:", min(scores))

First score: 72
Last score: 78
Number of scores: 5
Average: 78.0
Highest score: 91
Lowest score: 64
```

Exercise 3.2 — Lists change

A new member joins with a score of **88**, and you discover the 4th score (position 3) should be **66**, not 64. (a) Append 88 to the list, (b) replace position 3 with 66, (c) print the updated list and its new average.

```
[26]: scores.append(88)
      scores[3] = 66

print("Updated scores:", scores)
print("New average:", sum(scores) / len(scores))

Updated scores: [72, 85, 91, 66, 78, 88]
New average: 80.0
```

Part 4 · Dictionaries — Labeled Data

Exercise 4.1

Build a dictionary called `me` describing yourself with the keys "name", "age", "program", "district". Print your program using `me["program"]`.

```
[70]: me = {
      "name": "Osama",
      "age": 25,
      "program": "software engineering",
      "district": "Gasabo"
    }

    print (me["program"])

software engineering
```

Exercise 4.2 — Update and extend

(a) Add a new key "score" with a value of your choice, (b) increase "age" by 1, (c) print all the keys with `me.keys()` and then the whole dictionary

```
[32]: me["score"] = 90
      me["age"] = me["age"] + 1

      print (me.keys())
      print (me)

dict_keys(['name', 'age', 'program', 'district', 'score'])
{'name': 'Osama', 'age': 26, 'program': 'software engineering', 'district': 'Gasabo', 'score': 90}
```

Part 5 · Class Dataset Analysis

a list of 20 dictionaries called students.

```
[71]: students = [
      {"id": "AUCA001", "name": "Aline Uwase", "age": 21, "gender": "F", "program": "Information Systems", "district": "Gasabo", "attendance_pct": 92, "score": 85},
      {"id": "AUCA002", "name": "Eric Niyonzima", "age": 23, "gender": "M", "program": "Software Engineering", "district": "Kicukiro", "attendance_pct": 88, "score": 78},
      {"id": "AUCA003", "name": "Diane Mukamana", "age": 20, "gender": "F", "program": "Information Systems", "district": "Nyarugenge", "attendance_pct": 85, "score": 82},
      {"id": "AUCA004", "name": "Jean Bosco Habimana", "age": 24, "gender": "M", "program": "Networking", "district": "Gasabo", "attendance_pct": 70, "score": 75},
      {"id": "AUCA005", "name": "Clarisse Ingabire", "age": 22, "gender": "F", "program": "Accounting", "district": "Musanze", "attendance_pct": 85, "score": 80},
      {"id": "AUCA006", "name": "Patrick Mugisha", "age": 21, "gender": "M", "program": "Software Engineering", "district": "Huye", "attendance_pct": 90, "score": 88},
      {"id": "AUCA007", "name": "Sandrine Umutohi", "age": 19, "gender": "F", "program": "Information Systems", "district": "Kicukiro", "attendance_pct": 80, "score": 72},
      {"id": "AUCA008", "name": "Emmanuel Nsengiyumva", "age": 25, "gender": "M", "program": "Networking", "district": "Rubavu", "attendance_pct": 65, "score": 68},
      {"id": "AUCA009", "name": "Grace Uwimana", "age": 20, "gender": "F", "program": "Accounting", "district": "Gasabo", "attendance_pct": 82, "score": 85},
      {"id": "AUCA010", "name": "Yves Ndayisenga", "age": 22, "gender": "M", "program": "Software Engineering", "district": "Nyarugenge", "attendance_pct": 88, "score": 82},
      {"id": "AUCA011", "name": "Josiane Mukeshimana", "age": 21, "gender": "F", "program": "Information Systems", "district": "Muhanga", "attendance_pct": 78, "score": 75},
      {"id": "AUCA012", "name": "Claude Iradukunda", "age": 23, "gender": "M", "program": "Networking", "district": "Gasabo", "attendance_pct": 75, "score": 72},
      {"id": "AUCA013", "name": "Beata Nyirahabimana", "age": 24, "gender": "F", "program": "Accounting", "district": "Nyamagabe", "attendance_pct": 89, "score": 85},
      {"id": "AUCA014", "name": "Olivier Twagirimana", "age": 20, "gender": "M", "program": "Software Engineering", "district": "Kicukiro", "attendance_pct": 80, "score": 78},
      {"id": "AUCA015", "name": "Chantal Uwera", "age": 22, "gender": "F", "program": "Information Systems", "district": "Rusizi", "attendance_pct": 88, "score": 82},
      {"id": "AUCA016", "name": "Fabrice Byiringiro", "age": 21, "gender": "M", "program": "Networking", "district": "Musanze", "attendance_pct": 64, "score": 68},
      {"id": "AUCA017", "name": "Solange Ishimwe", "age": 19, "gender": "F", "program": "Software Engineering", "district": "Gasabo", "attendance_pct": 90, "score": 85},
      {"id": "AUCA018", "name": "David Hakizimana", "age": 25, "gender": "M", "program": "Accounting", "district": "Nyagatane", "attendance_pct": 72, "score": 75},
      {"id": "AUCA019", "name": "Immaculee Uwamahoro", "age": 23, "gender": "F", "program": "Information Systems", "district": "Kicukiro", "attendance_pct": 85, "score": 80},
      {"id": "AUCA020", "name": "Samuel Rukundo", "age": 20, "gender": "M", "program": "Networking", "district": "Nyarugenge", "attendance_pct": 94, "score": 88}
    ]

[40]: print(len(students))

20
```

```
names = [s["name"] for s in students] # every student's name
all_scores = [s["score"] for s in students] # every score
```

```
districts = [s["district"] for s in students] # every district
```

```
[45]: names = [s["name"] for s in students]
      all_scores = [s["score"] for s in students]
      districts = [s["district"] for s in students]

[46]: names = [s["name"] for s in students]
      all_scores = [s["score"] for s in students]
      districts = [s["district"] for s in students]

[48]: print(len(names))
      print(len(all_scores))
      print(len(districts))

      20
      20
      20
```

Exercise 5.1 — Describe the class

Using f-strings, print: (a) how many students are in the dataset, (b) the class average score rounded to 1 decimal — `round(..., 1)`, (c) the highest and lowest score, (d) the range (highest minus lowest)

```
[51]: print(f"Number of students: {len(students)}")
      print(f"Class average score: {round(sum(all_scores) / len(all_scores), 1)}")
      print(f"Highest score: {max(all_scores)}")
      print(f"Lowest score: {min(all_scores)}")
      print(f"Score range: {max(all_scores) - min(all_scores)}")

      Number of students: 20
      Class average score: 77.2
      Highest score: 94
      Lowest score: 55
      Score range: 39
```

Exercise 5.2 — Inspect individual records

(a) Print the full record of the **first** and the **last** student. (b) For the student at position 6, print one f-string sentence:
NAME from DISTRICT studies PROGRAM and scored SCORE.

```
[54]: print("First student:", students[0])
      print("Last student:", students[-1])

      student = students[6]

      print(f"{student['name']} from {student['district']} studies {student['program']} and scored {student['score']}")

      First student: {'id': 'AUCA001', 'name': 'Aline Uwase', 'age': 21, 'gender': 'F', 'program': 'Information Systems', 'district': 'Gasabo', 'attendance_pct': 92, 'score': 85}
      Last student: {'id': 'AUCA020', 'name': 'Samuel Rukundo', 'age': 20, 'gender': 'M', 'program': 'Networking', 'district': 'Nyarugenge', 'attendance_pct': 94, 'score': 92}
      Sandrine Umutoni from Kicukiro studies Information Systems and scored 94.
```

Exercise 5.3 — Count by category

Lists have a built-in counter: `districts.count("Gasabo")`. Print how many students come from (a) Gasabo, (b)

Kicukiro, and (c) one other district of your choice.

```
[59]: print ("Gasabo:", districts.count("Gasabo"))
      print("Kicukiro:", districts.count("Kicukiro"))
      print ("Musanze:", districts.count("Musanze"))

Gasabo: 5
Kicukiro: 4
Musanze: 2
```

Exercise 5.4 — Stretch goal: who is the top student?

Without any loop, print: *The top student is NAME with a score of SCORE.*

Combine three things you already know:

1. `max(all_scores)` → the best score
2. `all_scores.index(...)` → its position
3. `names[position]` → the name

```
[61]: top_score = max(all_scores)
      top_position = all_scores.index(top_score)
      top_name = names[top_position]

      print(f"The top student is {top_name} with a score of {top_score}.")

The top student is Sandrine Umutoni with a score of 94.

[63]: lowest_score = min(all_scores)
      lowest_position = all_scores.index(lowest_score)
      lowest_name = names[lowest_position]

      print(f"The lowest student is {lowest_name} with a score of {lowest_score}.")

The lowest student is David Hakizimana with a score of 55.

[64]: top_student = students[top_position]
      lowest_student = students[lowest_position]

      print(f"Top student: {top_student['name']} - Score: {top_student['score']} - Attendance: {top_student['attendance_pct']}%")
      print(f"Lowest student: {lowest_student['name']} - Score: {lowest_student['score']} - Attendance: {lowest_student['attendance_pct']}%")

Top student: Sandrine Umutoni - Score: 94 - Attendance: 97%
Lowest student: David Hakizimana - Score: 55 - Attendance: 72%
```

Part 6 · Reflection

1. Is this 20-student dataset Big Data?

No, I don't think this dataset is Big Data because it only contains 20 students. It is small and easy to work with on a normal computer. It does not clearly have the Big Data V's such as large Volume, high Velocity, and Variety.

2. What surprised or confused you?

One thing that surprised me was that "3" + "4" gives "34" instead of 7. At first, I expected the answer to be 7, but I learned that Python sees them as text because they are strings. This helped me understand why data types are important in Python.

Week 1 Lab Report — Python Foundations

In this week's lab, I learned some basic Python skills that I think will be very useful for Big Data Analytics. Since I am still a beginner in Python, the exercises helped me understand how Python works step by step.

I learned how to create variables and use different data types such as strings, integers, floats, and Boolean values. I also learned how to convert text into a number. Working with strings helped me understand how to change text, count characters, and get specific characters from a word or sentence.

I also practiced using lists and learned how to calculate an average, find the highest and lowest values, and update values in a list. After that, I learned about dictionaries and how they can be used to store information with labels such as name, age, program, and district.

The most useful part for me was analyzing the 20-student dataset. I was able to find the class average, highest and lowest scores, count students from different districts, and identify the top student. This made me understand how Python can be used to work with real data instead of only doing simple examples.

One thing that surprised me was that "3" + "4" gives "34" instead of 7. I learned that this happens because they are strings, not numbers. Overall, I found this lab very helpful because it gave me a simple introduction to Python and made me more comfortable working with data.